ECS601U Control Systems

Lab Report

Matlab and Simulink for Control Design and Simulation (Part 1 & 2)

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1 Lab 1

1.1 Introduction

Matlab is an interactive working environment for numerical computations and data visualisation that is extremely useful for control systems analysis and design. It provides industry-standard algorithms and apps for systematically analysing, design and tuning linear control systems. You can specify your system as a transfer function, state-space, pole-zero-gain, or frequency-response model. Apps and functions, such as step response plot and Bode plot, let you visualise system behaviour in time domain and frequency domain.

Simulink is a block diagram environment for interactive system modelling, analysis and simulation. It provides a graphical editor, customizable block libraries, and solvers for modelling and simulating control systems. Complex systems also containing nonlinearities can be built and analysed easily in graphical form. It is integrated with Matlab, enabling you to incorporate Matlab algorithms into models and export simulation results to Matlab for further analysis.

This lab is designed to help you grasp a better understanding of the theory and principles of control systems discussed in lectures.

Learning Objectives:
- To be familiar with representing polynomials in Matlab.
- To be able to find roots of polynomials and create polynomials when roots are known.
- To be able to obtain partial fraction expansion of the ratio of two polynomials in Matlab.
- To be able to reduce a linear system block diagram to its canonical form and find the equivalent transfer function in Matlab.
- To be able to compute and plot the time response of the LTI systems due to different input stimuli in Matlab.

1.2 Background Theory

Polynomials in Matlab

Matlab provides functions for standard polynomial operations, such as polynomials roots, evaluation, and differentiation. In addition, there are functions for more advanced applications, such as curve fitting and partial fraction expansion. Aside from the basic Matlab plotting commands, you should become familiar with the following commands:

roots - used to find roots of a polynomial.
Poly - used to create a polynomial whose roots are known.
Polyval - used to evaluate a polynomial for different input values.

Polyvalm - used to evaluate a polynomial with a matrix argument.
Conv - used to convolve two polynomials in t-domain, i.e. multiply two polynomials in s-domain.
Deconv - used to deconvolve two polynomials in t-domain, i.e. divide two polynomials in s-domain.
Polyder - used to differentiate a polynomial.
Polyfit - used to fit a polynomial to data, i.e. curve fitting.
Residue - used to obtain partial fraction expansion of the ratio of two polynomials.

Representing Polynomials in Matlab

Matlab represents polynomials as row vectors containing coefficients ordered by descending powers. For example, consider the equation: $p(x) = x^3 - 2x - 5$

Convolution and Deconvolution

Polynomial multiplication and division correspond to the operations convolution and deconvolution. The function **conv** and **deconv** implement these operations.

Step Response

Step response is the time behaviour of the outputs of a general system when its inputs change from zero to one in a very short time. To plot the unit-step response of the LTI system described bySYS = tf (num, den) in Matlab, use the function step (SYS). Alternatively, the row vectors of the numerator and denominator coefficients of the transfer function can be entered directly into the step function: step (num, den)

Impulse Response

The unit-impulse response of a control system described by **SYS = tf (num, den)** in Matlab can be plotted using the function **impulse (SYS)**

Ramp Response

There is no ramp function or command in Matlab. To obtain the unit ramp response of the transfer function G(s), multiply G(s) by $1/s$ and use the **step** function to plot the ramp response. The **step** function will further multiply the transfer function by $1/s$ to make the input $1/s^2$ i.e. Laplace transform of a unit ramp function.

Arbitrary Response

To obtain the time response of a LTI system described by SYS = t f (num, den) to an arbitrary input (e.g. exponential function, sinusoidal function, etc.), the lsim function (stands for 'linear simulation') can be used as follows:

```
>> lsim(SYS, r, t)
```
or `lsim(num, den, r, t)`

where num and den are the row vectors of the numerator and denominator coefficients of

the transfer function, r is the input time function, and t is the time range over which r is defined.

1.3 Results and Discussion

1) Calculate the following by hand and validate using Matlab:

By hand	Matlab
a) $p(s) = s^2 + 2s + 1$ $= (s+1)(s+1)$ $s = -1$, $s = -1$	p = [1 2 1]; r = roots (p) r = -1 -1
b) $q(s) = s+1$ $s = -1$	q = [1 1]; r = roots (q) r = -1
c) Convolution sum p(s) and q(s) y[0] = 1*1 = 1 y[1] = (1*1) + (2*1) = 3 y[2] = (1*0) + (2*1) + (1*1) = 3 y[3] = (1*0) + (2*0) + (1*1) + (0*1) = 1	p = [1 2 1]; q = [1 1]; c = conv (p, q) c = 1 3 3 1
d) $p(s) = s^2 + 2s + 1$ $p(-1) = (-1)^2 + 2(-1) + 1 = 0$ $q(6) = 6 + 1 = 7$	polyval (p, -1) ans = 0 polyval (q, 6) ans = 7

2) Calculate by hand the partial fraction expansion of the following and validate using Matlab:

By Hand	Matlab
$G(s) = \frac{5(s+2)}{s(s^2+8s+15)}$ $5(s+2) = \frac{A}{s} + \frac{B}{s+5} \frac{C}{s+3}$ $5(s+2) = A(s+5)(s+3) + B(s)(s+3) + C(s)(s+5)$ $s = -5$ $5(-5+2) = 0 + B(-5)(-2) + 0$ $-15 = B(10)$ $B = -3/2$ $s = -3$ $5(-3+2) = 0 + 0 + C(-3)(2)$ $C = 5/6$ $s = 0$ $5(0+2) = A(15) + 0 + 0$ $10 = A(15)$ $A = 10/15$ $A = 2/3$	>> b = [5 10] b = 5 10 >> a = [1 8 15 0] a = 1 8 15 0 >> [r, p, k] = residue(b,a) r = -1.5000 0.8333 0.6667 p = -5 -3 0 k = []

By Hand	Matlab
$G(s) = \frac{s^2+2s+3}{(s+1)^3}$ $G(s) = \frac{s^2+2s+3}{(s+1)^3} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{(s+1)^3}$ $s^2+2s+3 = A(s+1)^2 + B(s+1) + C$ $s^2+2s+3 = A(s^2+2s+1) + B(s+1) + c$ We can derive A, B and C as: $s^2 = As^2 + 0 \quad therefore\ A = 1.$ $2 = 2A + B + 0 \ \ Substitute\ A = 1\ gives\ B = 0$ $3 = A1 + B + C\ Substitute\ values\ of\ A\ and\ B\ gives\ C = 2.$ $G(s) = \frac{s^2+2s+3}{(s+1)^3} = \frac{1}{s+1} + \frac{0}{(s+1)^2} + \frac{2}{(s+1)^3}$	b = 1 2 3 >> a = [1 3 3 1] a = 1 3 3 1 >> [r, p, k]=residue(b,a) r = 1.0000 -0.0000 2.0000 p = -1.0000 -1.0000 -1.0000 k = []

3) Evaluate by hand and validate using Matlab the transfer function of the feedback system shown.

Polynomials are shown in Matlab with the use of vectors which contains coefficients of each power of the polynomial with the highest power in the lead. For example, consider the polynomial. $p(x) = s^3 + 2s^2 + 5s$ This can be entered in Matlab using the coefficients as:

```
>> p = [1 2 5 0];
```

We have 0 because the polynomial does not contain the last constant term, so for every term that is not present in the polynomial we represent its coefficient with zero.

The roots of a known polynomial can be calculated using the roots function, for our polynomial p, the roots are found using the following Matlab code.

```
>> r = roots(p)
```

A transfer function is described as the relationship between the output and input of a system, this is described using the ratio of polynomials in Matlab. This can be entered in Matlab using the command tf (num,den). Where num and den are vectors containing coefficients of the numerator and the denominator polynomials of the transfer function.

One of the shortcomings of using Matlab to evaluate partial fraction expansion is that Matlab cannot evaluate the roots of negative powers.

4) **Considering the following system: a- Use Matlab to find T(s) = b(s)/a(s), the closed-loop transfer function. Make use of the commands tf, conv, and feedback.**

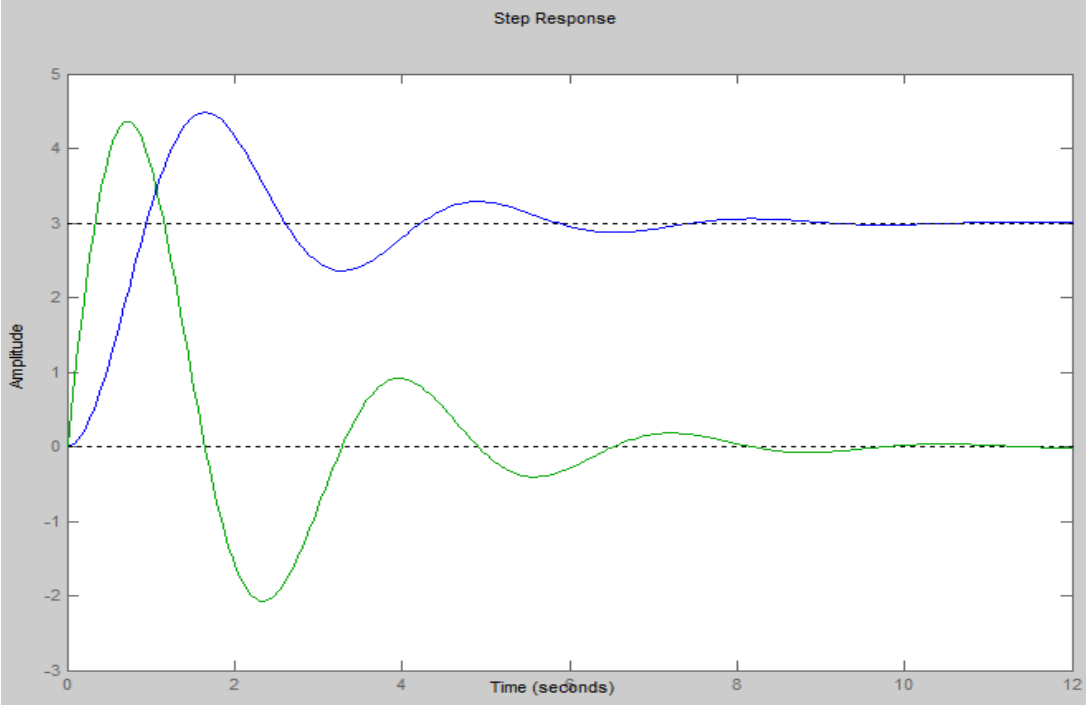
By hand	Matlab
$G1\ (s) = \frac{1}{s+1}$ $\quad G2\ (s) = \frac{s+2}{s+3}$ Multiply G1(s) * G2(s) $= \frac{1}{(s+1)} * \frac{s+2}{(s+3)} = \frac{s+2}{(s+1)(s+3)}$ $\frac{output}{input} = \frac{forward\ path\ gain}{1 \pm Loop\ path\ gain}$ $= \frac{G_1(s)G_2(s)}{1+G_1(s)G_2(s)H}$ $= \frac{s+2/(s+1)(s+3)}{1+H(s+2/(s+1)(s+3))}$ $= \frac{S+2}{S^2+5S+5}$	G1 = tf ([0 1], [1 1]); G2 = tf ([1 2] , [1 3]); H = tf ([0 1] , [0 1]); SYS = feedback (G1 * G2, H) SYS = $\frac{S+2}{S^2+5S+5}$

5a.

```
>> a = conv ([0 10], [1 2])
a =
    0    10    20
>>b = conv ([1 -1], [1 3])
b =
    1     2    -3
c = conv (b, [1 0])
c =
    1     2    -3     0
>> G = tf (a, c)
G = 10 s + 20
  -----------------
  s^3 + 2 s^2 - 3 s
Continuous-time transfer function
>> d = conv ([0 1], [1 1]);
>> e = conv ([0 1], [1 3]);
>> H = tf (d, e);
>> SYS = feedback (G, H)
SYS =
        10 s^2 + 50 s + 60
```

```
  --------------------------------
  s^4 + 5 s^3 + 13 s^2 + 21 s + 20
Continuous-time transfer function
```

5b. >> step(SYS)
>> hold on
>> impulse(SYS)



Plot shows the unit step response and the unit-impulse response of the system. The blue line shows that the unit- step response has a higher amplitude and an average value of 3. The green line shows the impulse response has an average value of 0.

5c. K = 10
>> f = 10 *G

```
ans =
   100 s + 200
  -----------------
  s^3 + 2 s^2 - 3 s
Continuous-time transfer function
```

>> SYS = feedback (f, H)

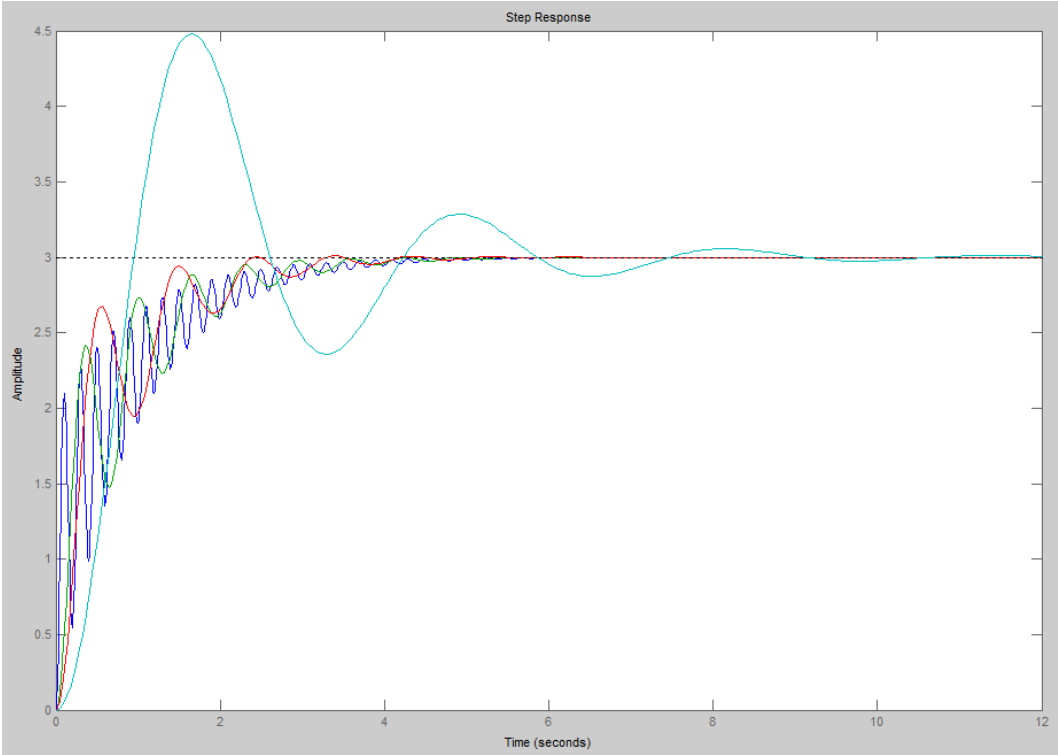
```
SYS =
         100 s^2 + 500 s + 600
  -------------------------------------
  s^4 + 5 s^3 + 103 s^2 + 291 s + 200
```

5d. I = 1*G

```
I =
     10 s + 20
  -----------------
  s^3 + 2 s^2 - 3 s
Continuous-time transfer function

>> SYS = feedback (I, H)
SYS =
         50 s^2 + 250 s + 300
  -----------------------------------
  s^4 + 5 s^3 + 53 s^2 + 141 s + 100
 Continuous-time transfer function
>> step (SYS)
>> hold on
>> J = 5*G;
>> SYS=feedback (J, H)
SYS =
         50 s^2 + 250 s + 300
  -----------------------------------
 s^4 + 5 s^3 + 53 s^2 + 141 s + 100
 Continuous-time transfer function
>> step (SYS)
>> hold on
>> k = 10*G;
>> SYS = feedback (k, H)
SYS =
         100 s^2 + 500 s + 600
  -----------------------------------
  s^4 + 5 s^3 + 103 s^2 + 291 s + 200
Continuous-time transfer function
>> step (SYS)
>> hold on
>> L = 100*G;
>> SYS = feedback (L, H)
SYS = 1000 s^2 + 5000 s + 6000
    ----------------------------------------------
  s^4 + 5 s^3 + 1003 s^2 + 2991 s + 2000
 Continuous-time transfer function
>> step (SYS)
```

The plot shows that as the gain constant (K) increases the frequency increases as well but the amplitude decreases.



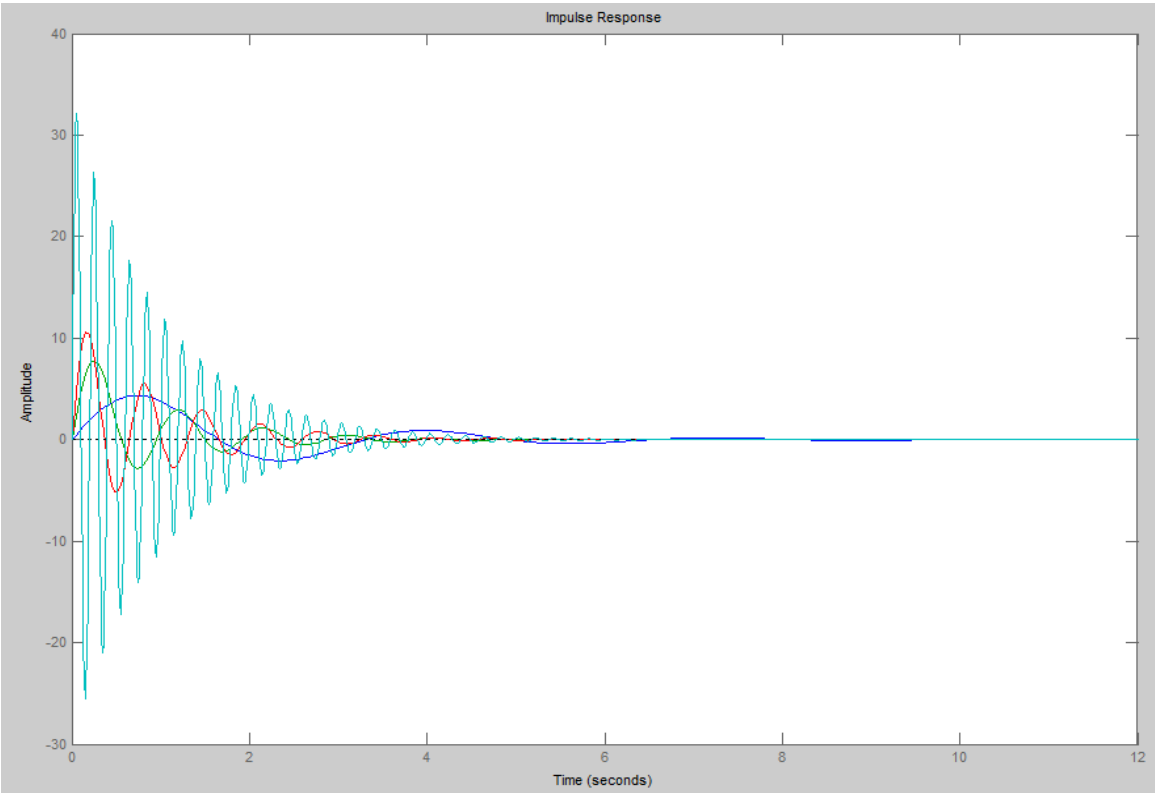
5e.

```
I = 1*G
I =
     10 s + 20
  -----------------
  s^3 + 2 s^2 - 3 s
 Continuous-time transfer function

>> SYS = feedback (I, H)
SYS =
         50 s^2 + 250 s + 300
  ----------------------------------
  s^4 + 5 s^3 + 53 s^2 + 141 s + 100
 Continuous-time transfer function
>> impulse (SYS)
>> hold on
>> J = 5*G;
>> SYS=feedback (J, H)
SYS =
         50 s^2 + 250 s + 300
  ----------------------------------
 s^4 + 5 s^3 + 53 s^2 + 141 s + 100
 Continuous-time transfer function
>> impulse (SYS)
```

```
>> hold on
>> k = 10*G;
>> SYS = feedback (k, H)
SYS =
        100 s^2 + 500 s + 600
  -----------------------------------
  s^4 + 5 s^3 + 103 s^2 + 291 s + 200
Continuous-time transfer function
>> impulse (SYS)
>> hold on
>> L = 100*G;
>> SYS = feedback (L, H)
SYS = 1000 s^2 + 5000 s + 6000
 ---------------------------------------------------
  s^4 + 5 s^3 + 1003 s^2 + 2991 s + 2000
 Continuous-time transfer function
>> impulse (SYS)
```

The plot shows that as the gain constant k increases, the frequency and the amplitude increase.



1.4 Summary

We have seen the capacity of Matlab as an interactive working environment and how some of these algorithms and applications are essential programs which a Control Engineer can use for analysis and design. This lab report has given us a better insight of the calculations by hand and those by Matlab. Simulink was used for modelling, simulating and analysing system unit-step and impulse responses.

2 Lab 2

2.1 Introduction

After the control engineer obtains a mathematical representation of a system (or subsystem), the system (or subsystem) is analysed for its transient and steady-state responses to see if these characteristics yield the desired behaviour. This lab is devoted to the analysis of system transient response. With the help of two examples, a RC circuit and a DC motor, the use of Matlab and Simulink for modelling and analysis of first-order and second-order systems is demonstrated.

Learning Objectives:
- To use Matlab and Simulink to determine the time response of a control system.
- Describe quantitatively the transient response of first-order and second-order systems.
- To find the damping ratio and natural frequency of a second-order system.
- Find the settling time, peak time, percent overshoot, and rise time for an under-damped second-order system.

2.2 Background Theory

First System Order

Consider a simple series combination of two passive electrical elements: a resistor, and a capacitor, known as a RC circuit, shown as:

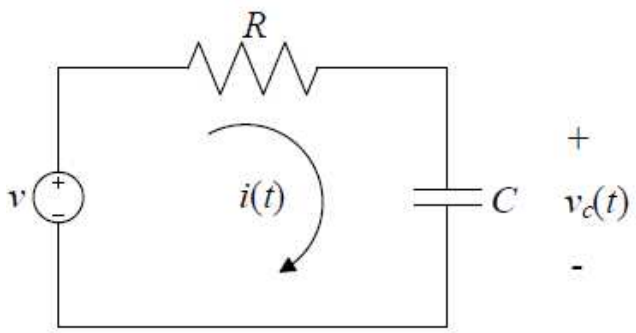


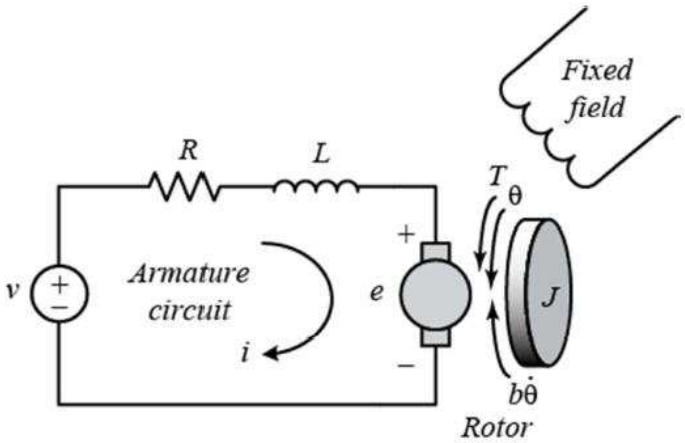
Figure 2-1: RC Circuit

The input is the voltage v in Volts and the output is the voltage across the capacitor $v_c(t)$. Since this circuit is a single loop, each node has one input and output; therefore, application of Kirchhoff's current law simply shows that the current is the same throughout the circuit at any given time. Now summing the voltages around the loop, assuming zero initial conditions, yields the system equation for this network as:

Refer to Lab sheet 1 and 2 for System Equations and Transfer Function

Second System Order

Consider a DC motor, whose electric circuit of the armature and the free body diagram of the rotor:



The rotor and the shaft are assumed to be rigid. The input is the armature voltage v in Volts (driven by a voltage source). Measured variables are the angular velocity of the shaft $\dot{\theta}$ in radians per second, and the shaft angle θ in radians. Consider the following values for the physical parameters:

Moment of inertia of the rotor J = 0.01 kg.m2

Damping (viscous friction) of the mechanical system b = 0.1 Nms

Back electromotive force constant K = 0.01 Nm/A

Electric resistance R = 1 Ω

Electric inductance L = 0.5 H

Using Simulink

Load Simulink by simply typing `simulink` in the Matlab command window. Please note that all graphics are from a PC running Matlab version R2012b and the views may differ on other versions of Matlab. Moreover, note that a succession of menu selections in the following sections of this lab sheet will be separated by the > symbol, for example, to create a new model in Simulink click File > New > Model from the toolbar in Simulink library browser window.

2.3 <u>Results and Discussion</u>

The convolution sum of p(s) and q(s)

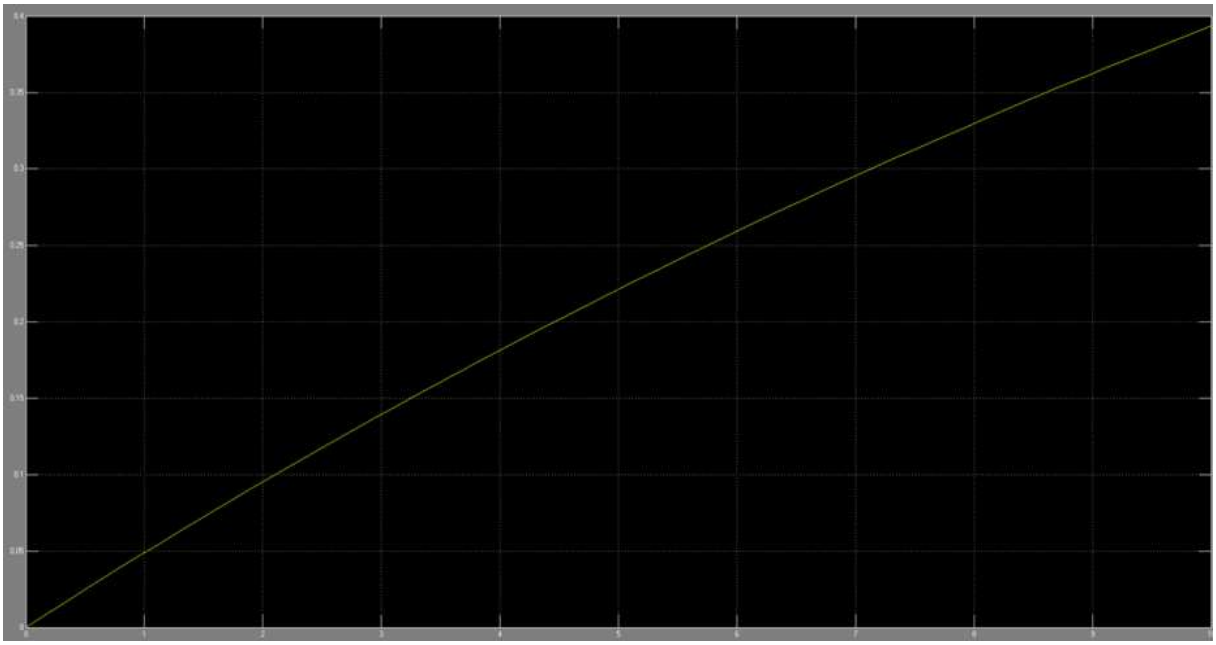
By Hand:

$p(s) = s^2 + 2s + 1$

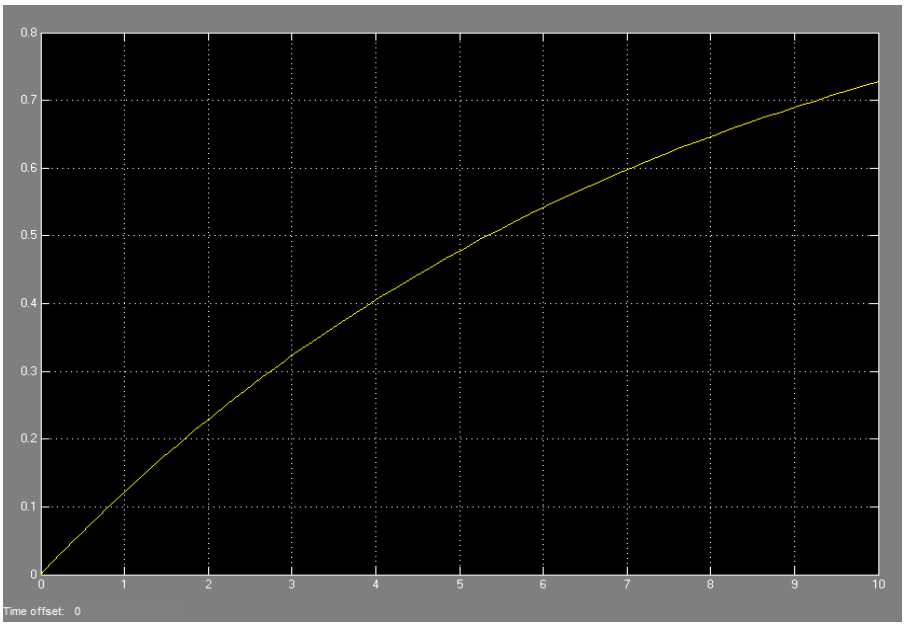
$q(s) = s + 1$

p(s) can be written in terms of co-efficients as:
[1 2 1]
Let the co-efficients of p(s) be represented as p_0, p_1, p_2
Similarly, q(s) can be written in terms of co-efficients as:
[1 1]
Let the co-efficients of p(s) be represented as q_0, q_1
Let output be y[s], so the convolution sum of p(s) and q(s) can be represented as:
y[0] = p[0] * q[0] = 1 * 1 = $\boxed{1}$
y[1] = (p[0] * q[1]) + (p[1] * q[0]) = (1 * 1) + (1 * 2) = $\boxed{3}$
y[2] = (p[0] * q[2]) + (p[1] * q[1]) + (p[2] * q[0]) = (1 * 0) + (2 * 1) + (1 * 1) = $\boxed{3}$
y[3] = (p[0] * q[3]) + (p[1] * q[2]) + (p[2] * q[1]) + (p[3] * q[0]) = (1 * 0) + (2 * 0) + (1 * 1) + (0 * 1) = $\boxed{1}$

$\boxed{\text{Q1a}}$
R = 2 kΩ and C = 0.01 F
τ =RC
τ =2000*0.01=20 s
Gain=1/ τ =1/20=0.05

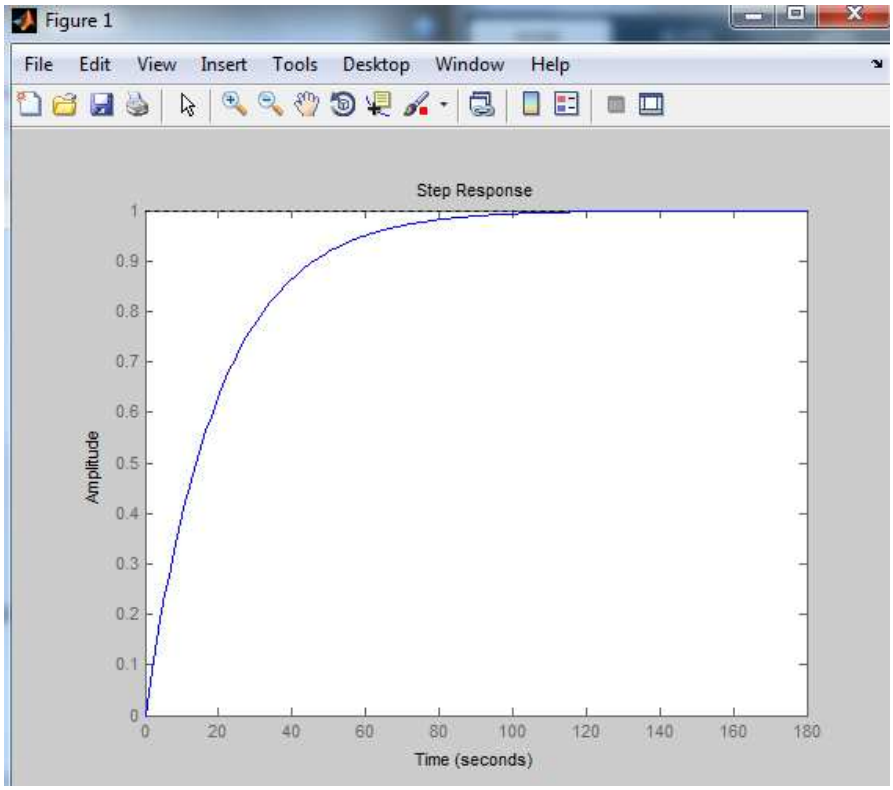


$\boxed{\text{Q1b}}$
R = 2.5 kΩ and C = 0.003 F
τ=RC
τ=2500*0.003=7.5 s
Gain=1/τ=1/7.5=0.13



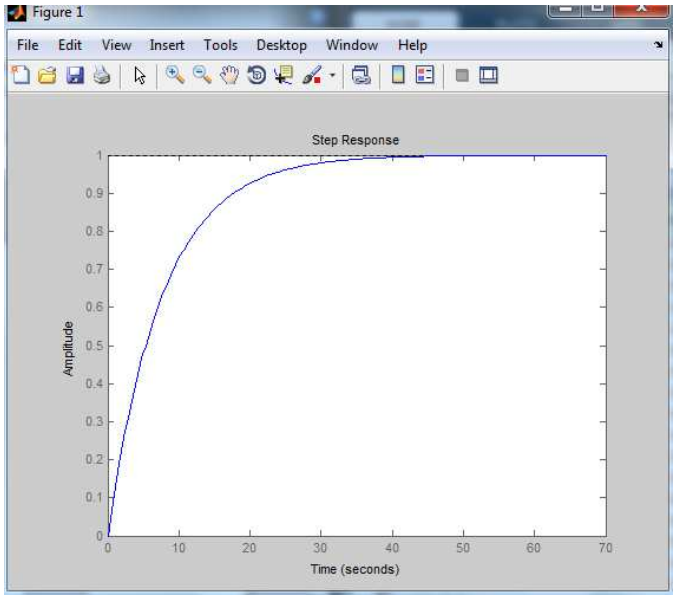
Q2a

```
>> num = [0 0.05];
>> den = [1 0.05];
>> SYS = tf (num, den)
SYS =
   0.05
  --------
  s + 0.05
 Continuous-time transfer function
>> step (SYS)
```



Q2b

```
>> num = [0 0.13];
>> den = [1 0.13];
>> SYS = tf(num, den);
>> step(SYS)
```



Q3a

$\omega_n = 1$ and $\zeta = 0.1, 0.4, 0.7, 1.0$ and 2.0

$\zeta = 0.1$ $\quad G(s) = \frac{1^2}{s(s+2(0.1)1)} = \frac{1^2}{s^2+0.2s}$

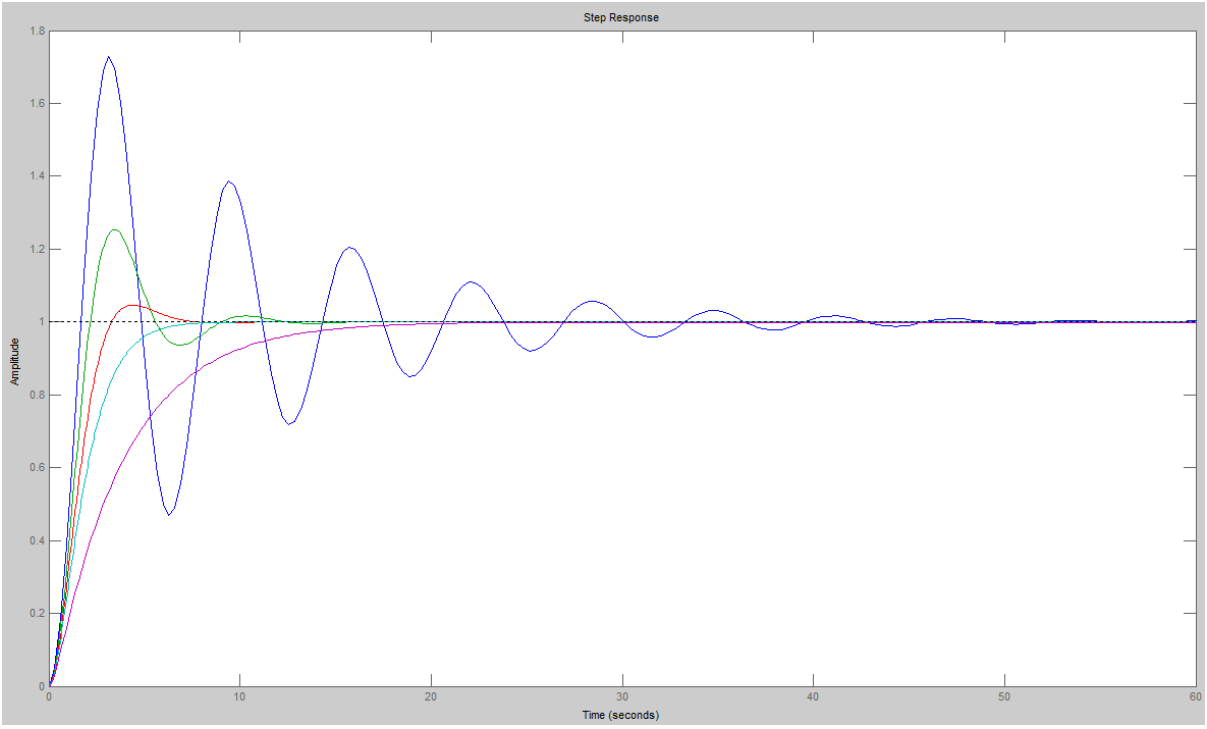
$\zeta = 0.4$ $\quad G(s) = \frac{1^2}{s(s+2(0.4)1)} = \frac{1^2}{s^2+0.8s}$

$\zeta = 0.7$ $\quad G(s) = \frac{1^2}{s(s+2(0.7)1)} = \frac{1^2}{s^2+1.4s}$

$\zeta = 1.0$ $\quad G(s) = \frac{1^2}{s(s+2(1.0)1)} = \frac{1^2}{s^2+2s}$

$\zeta = 2.0$ $\quad G(s) = \frac{1^2}{s(s+2(2.0)1)} = \frac{1^2}{s^2+4s}$

```
>> t1 = tf ([0 1], [1 0.2 0]);
>> H = tf ([0 1], [0 1]);
>> SYS = feedback (t1, H);
 >> t2 = tf ([0 1], [1 0.8 0]);
>> SYS1 = feedback (t2, H);
>> t3 = tf ([0 1], [1 1.4 0]);
>> SYS2 = feedback (t3, H);
>> t4 = tf ([0 1], [1 2 0]);
>> SYS3 = feedback (t4, H);
>> t5 = tf ([0 1], [1 4 0]);
>> SYS4 = feedback (t5, H);
>> step (SYS)
>> hold on
>> step (SYS1)
>> step (SYS2)
>> step (SYS3)
>> step (SYS4)
```



Q3b

$$T_r = \frac{\pi - \theta}{W_n} = \frac{\pi - \tan^{-1}\left(\sqrt{1 - \frac{\xi^2}{\xi}}\right)}{W_n\sqrt{1-\xi^2}}$$

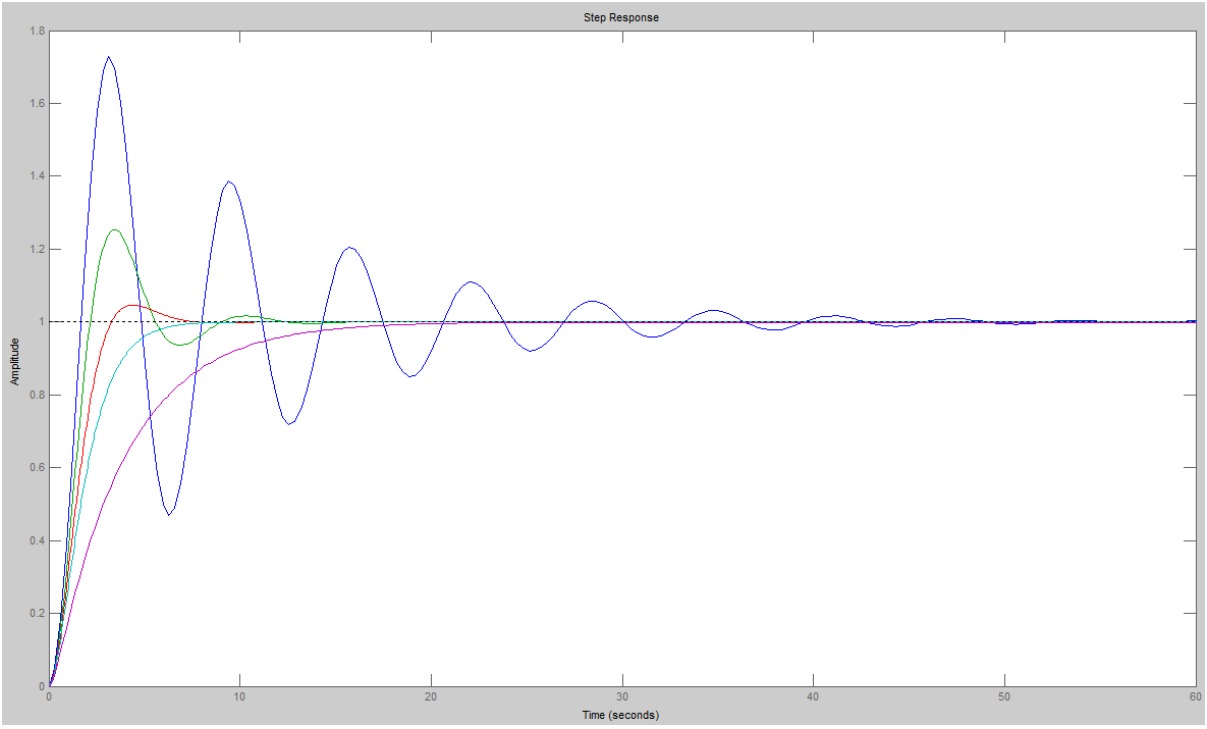
$$T_p = \frac{\pi}{W_n\sqrt{1-\xi^2}}$$

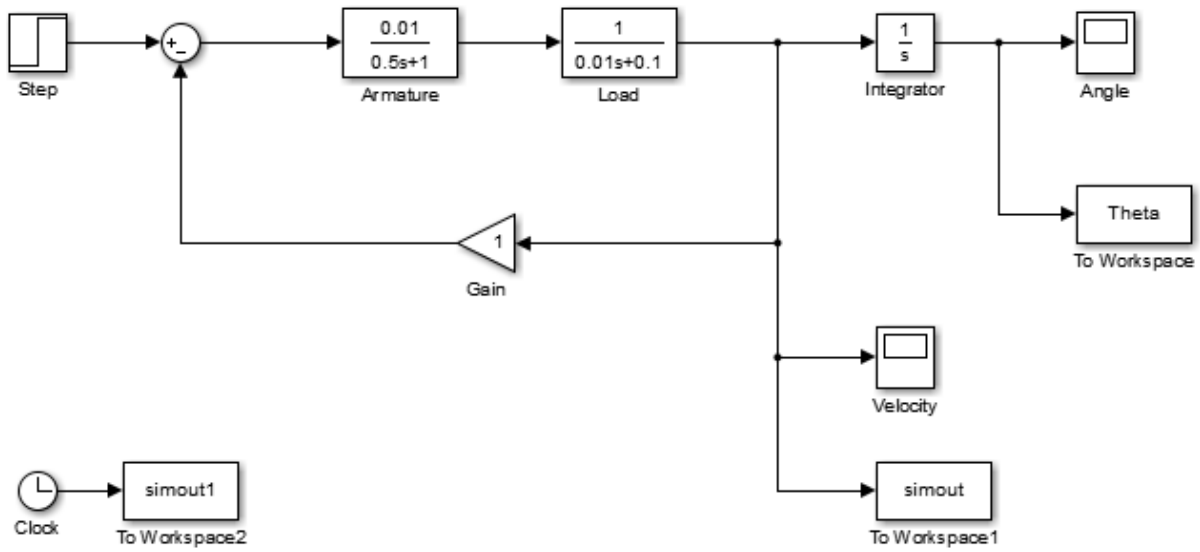
$$O/S = 100e^{\left(\frac{\xi\pi}{\sqrt{1-\xi^2}}\right)}$$

$$T_s = \frac{4}{\xi\, W_n}$$

Steady- State value = 1

ζ	Rise time Tr	Peak time Tp	% overshoot %OS	Settling time Ts	Steady-state value
0.1	1	3.24s	77%	40s	1
0.4	2.5s	3.95s	28%	10s	1
0.7	3s	4.74s	8%	5.71s	1
1.0	3.3	7.5	N/A	4	1
2.0	N/A	N/A	N/A	N/A	N/A

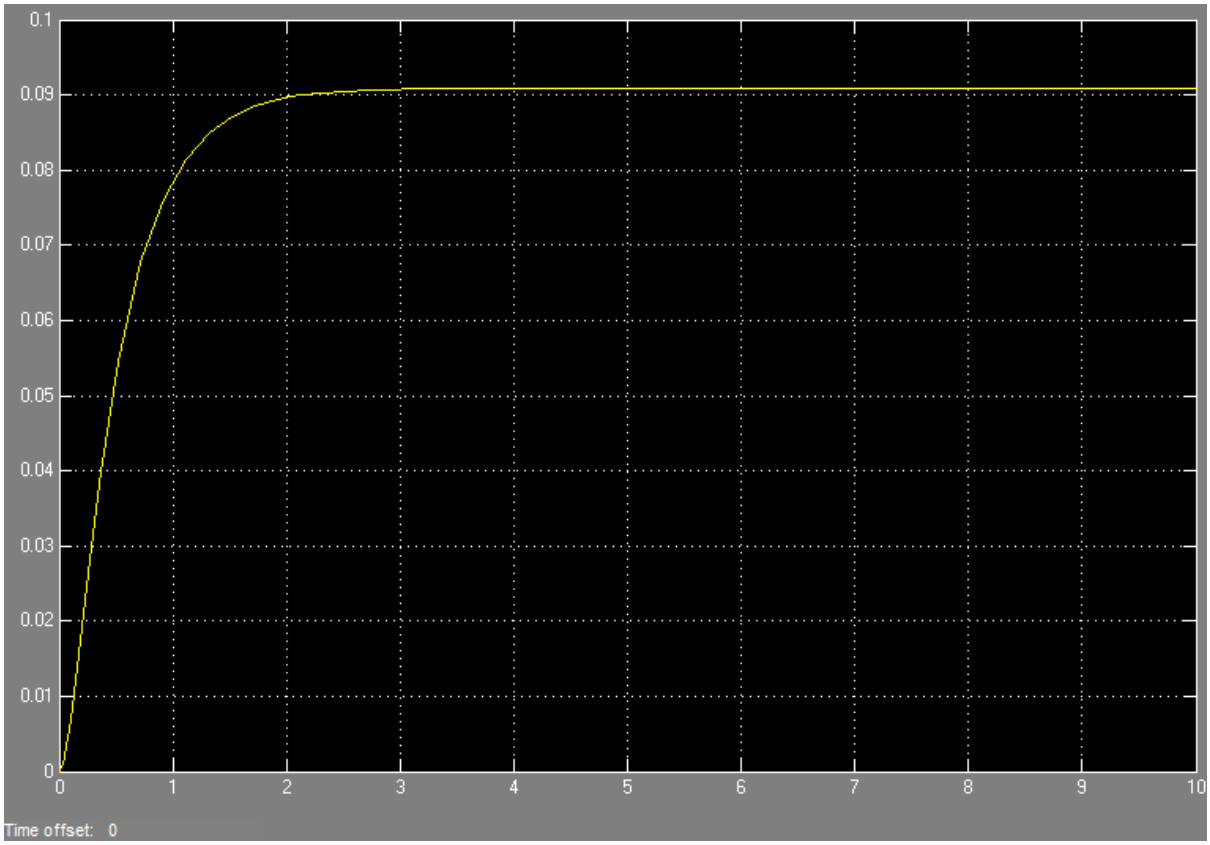




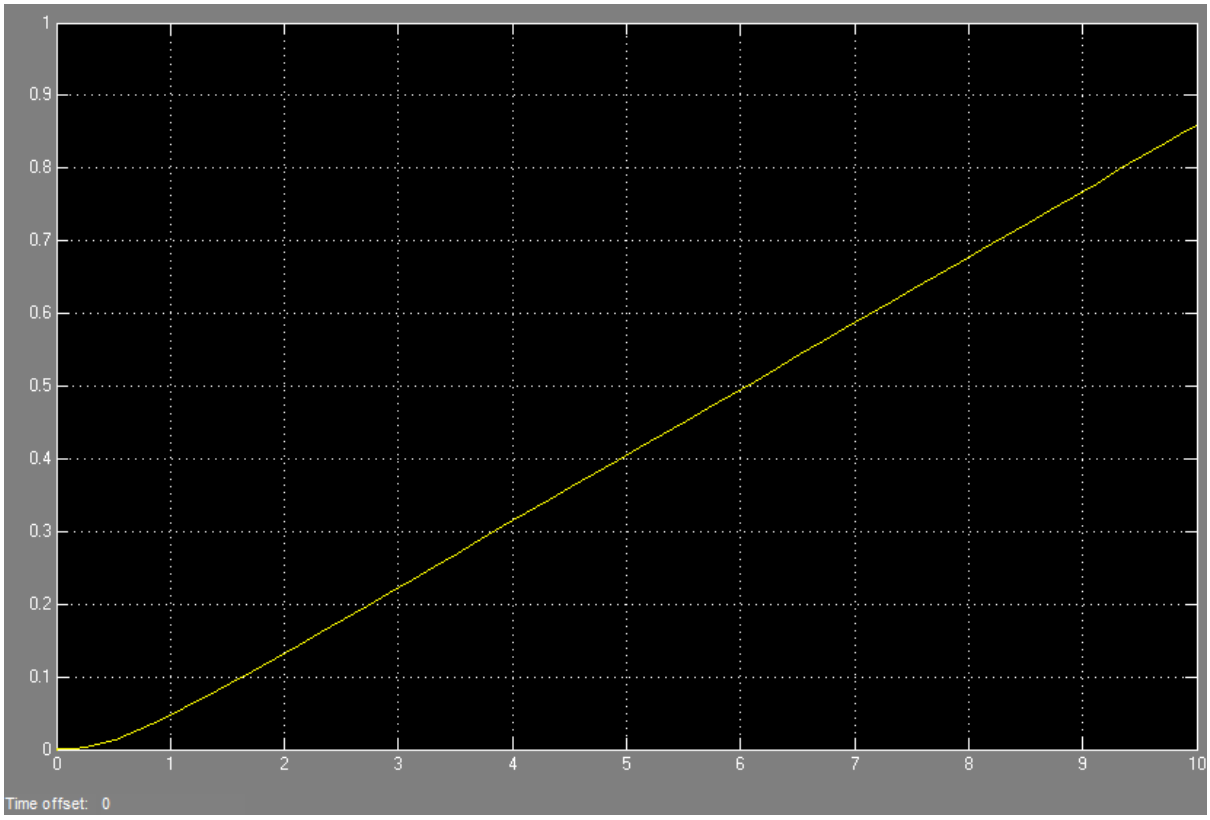
For **t=0,**

$\cos(2\pi t) = 1$

The time response of the DC motor **velocity** is:



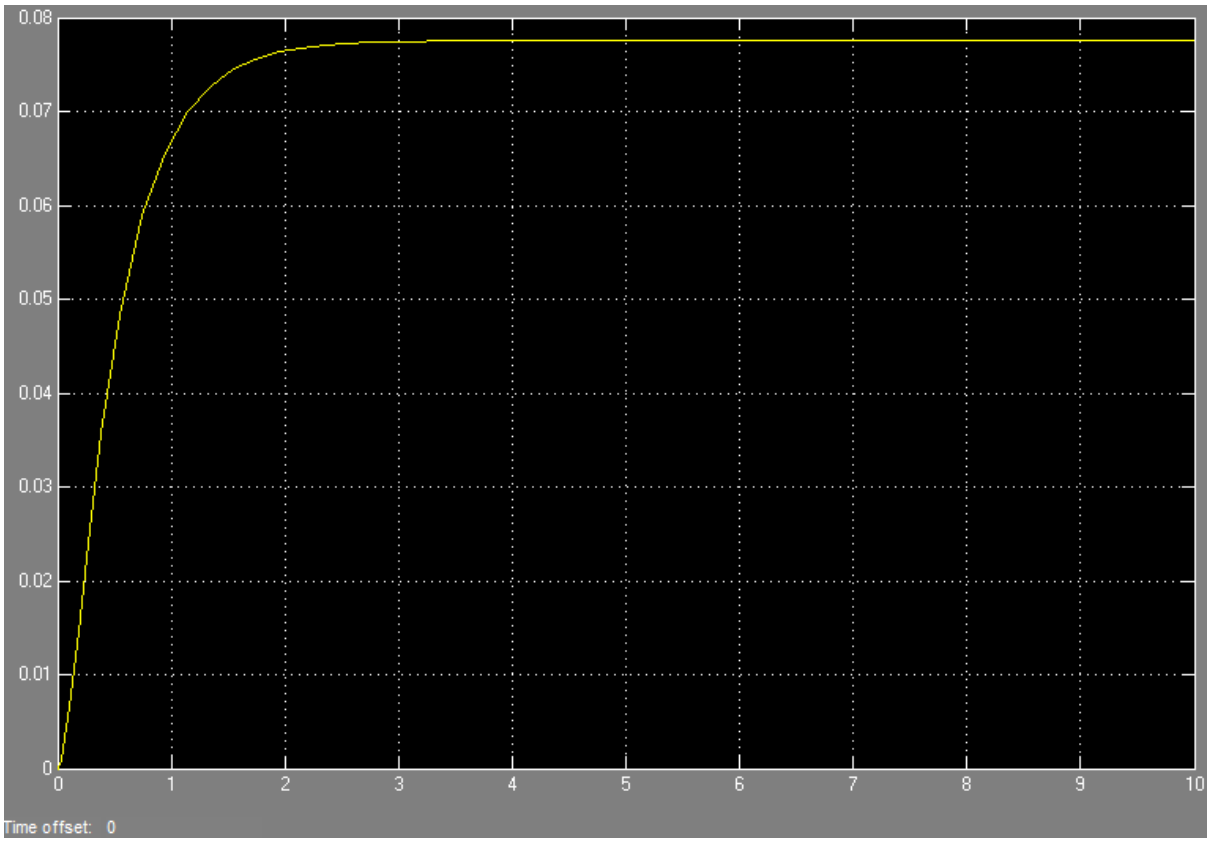
The time response of the DC motor **angle** is:



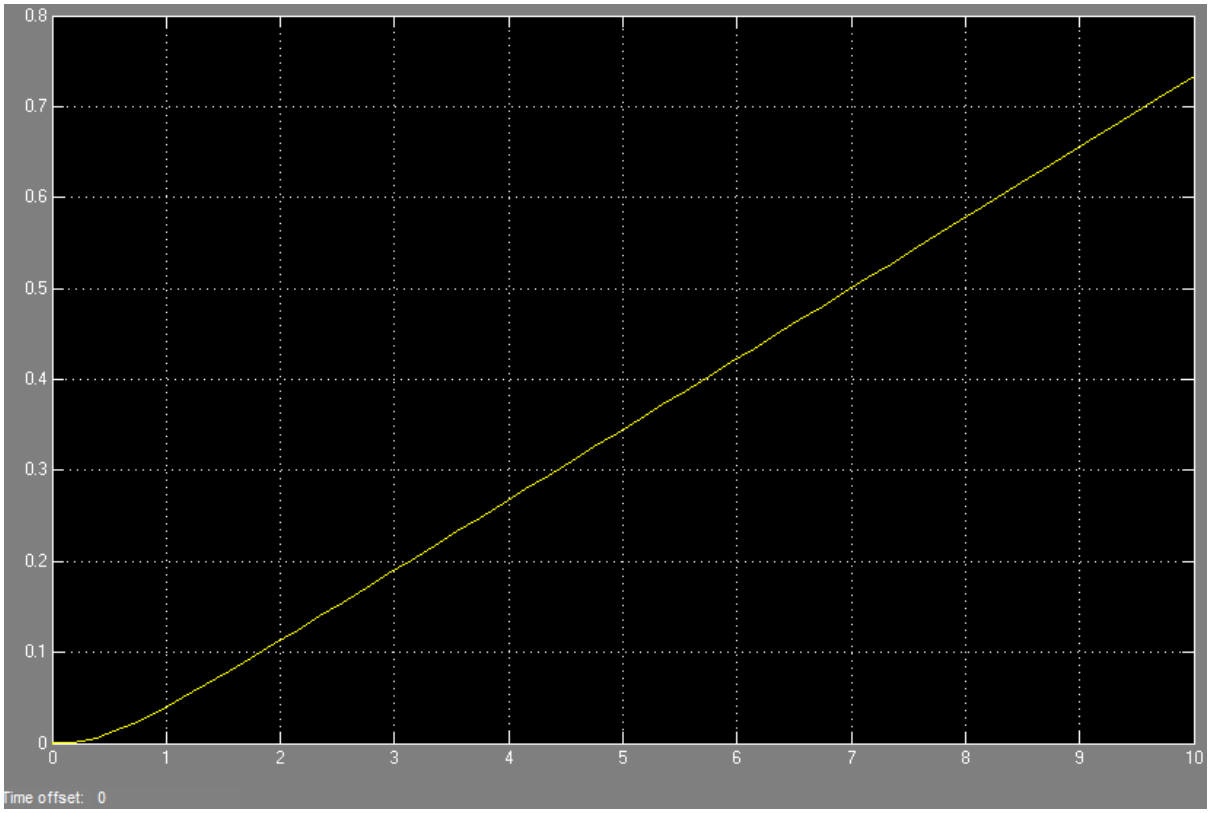
For **t=5,**

$\cos(2\pi t) = 0.853$

The time response of the DC motor **velocity** is:



The time response of the DC motor **angle** is:



2.4 Summary

We have learnt the nature of first and second order systems. An electric element connected in series was used to describe the first-order system and a DC motor was used to describe the second order system. We have compared and seen the behaviour of both systems to a unit step function. We can therefore conclude that both systems have a steady state value 1. However, the second order time response has much higher amplitude.

References:

- http://en.wikipedia.org/wiki/Control_system
- http://en.wikipedia.org/wiki/Step_response
- http://qmplus.qmul.ac.uk/mod/resource/view.php?id=266010
- http://qmplus.qmul.ac.uk/mod/resource/view.php?id=268361